

Mechanical Engineering Department

Detailed Die Design & Synthesis

Phase 3 Engineering Project

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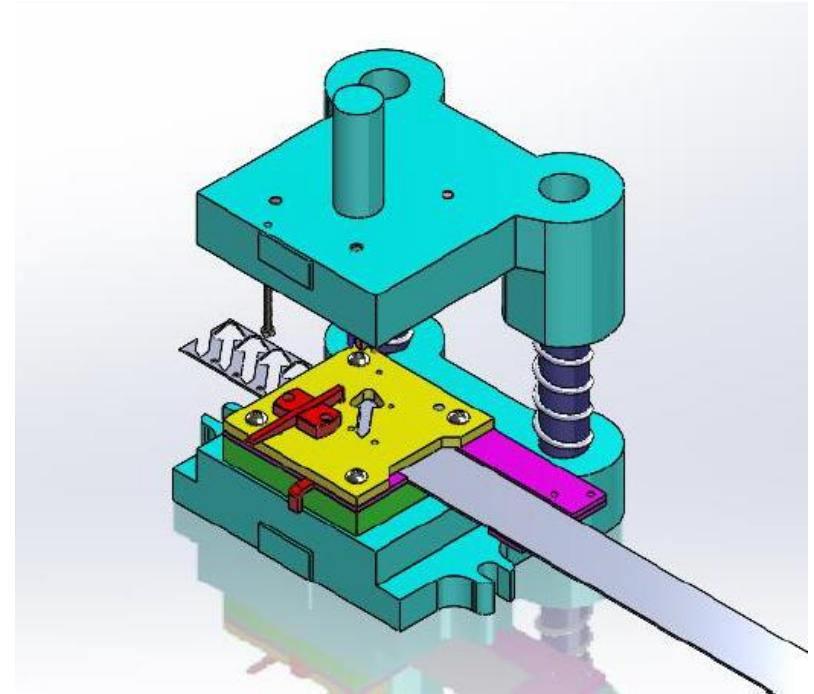
1. Project Definition & Goals: 3D CAD modeling in SolidWorks and industrial standards[cite: 14].

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5. Die Standard Components: FIBRO die set, matrix block, punches, stops, and gages[cite: 14].



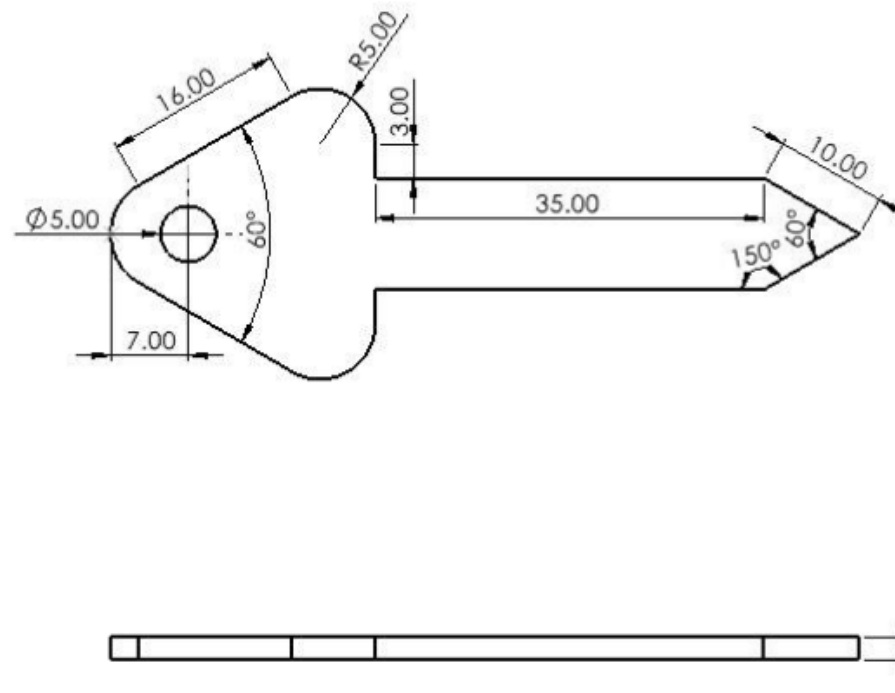
Project Definition & Objectives

PHASE 3 SCOPE

- **Full 3D CAD Modeling:** Shearing press die engineered in **SolidWorks** based on course principles.
- **Standard Components:** Incorporation of die set, matrix, blanking & piercing punches, stripper, and stops.
- **Manufacturing Documentation:** Detailed manufacturing drawings, BOM, and exploded assembly diagrams.
- **Catalog Adherence:** Standardization via **FIBRO**, **DIN**, and **ASME** catalogs.

Target Component Drawing

PART DIMENSIONS & GEOMETRY



Thickness: 2.00 mm

Hole: $\varnothing 5.00$ mm

Shank: 35.00 mm

Radius: R5.00 mm

1. Raw Material: St12 cold-rolled steel sheet, thickness **2.00 mm**, format **1000 × 1200 mm**

2. Configuration: Single Row – One Pass progressive shearing die.

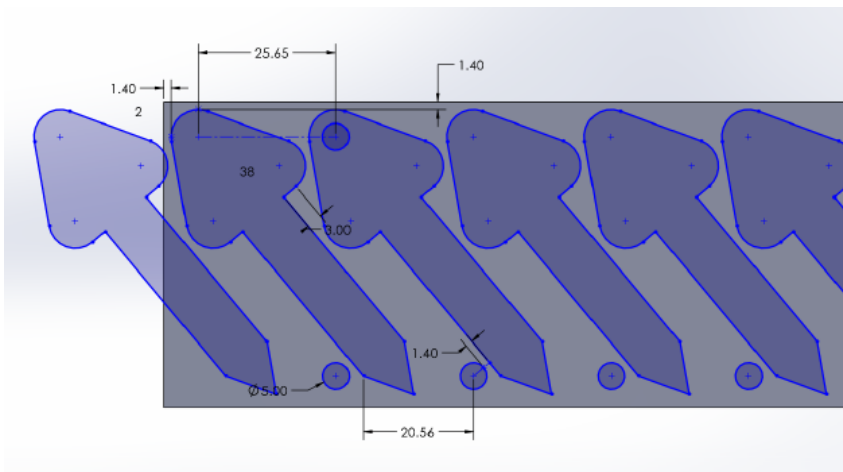
3. Strip Width: Slit via rollers to calibrated width of **56.79 mm**

- **Surface Quality:** Scale-free cold-rolled surface finish with high dimensional consistency.
- **Mechanical Properties:** Balanced ductility and shear resistance for clean blanking.
- **Supply Chain:** Delivered as coils, sheared to plates, and roller-slit to 56.79 mm feed strips.



Optimum Strip Nesting Layout

SCRAP REDUCTION



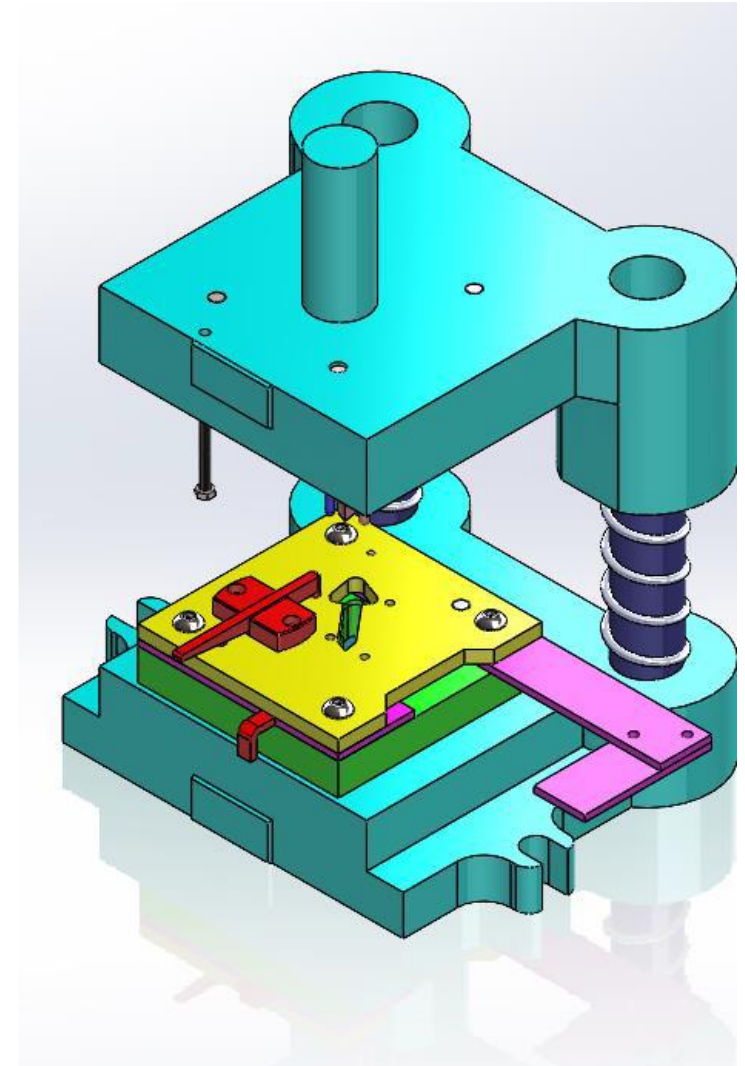
Angle: **-50.0°**
Pitch: **25.65 mm**
Edge/Bridge: **1.40 mm**

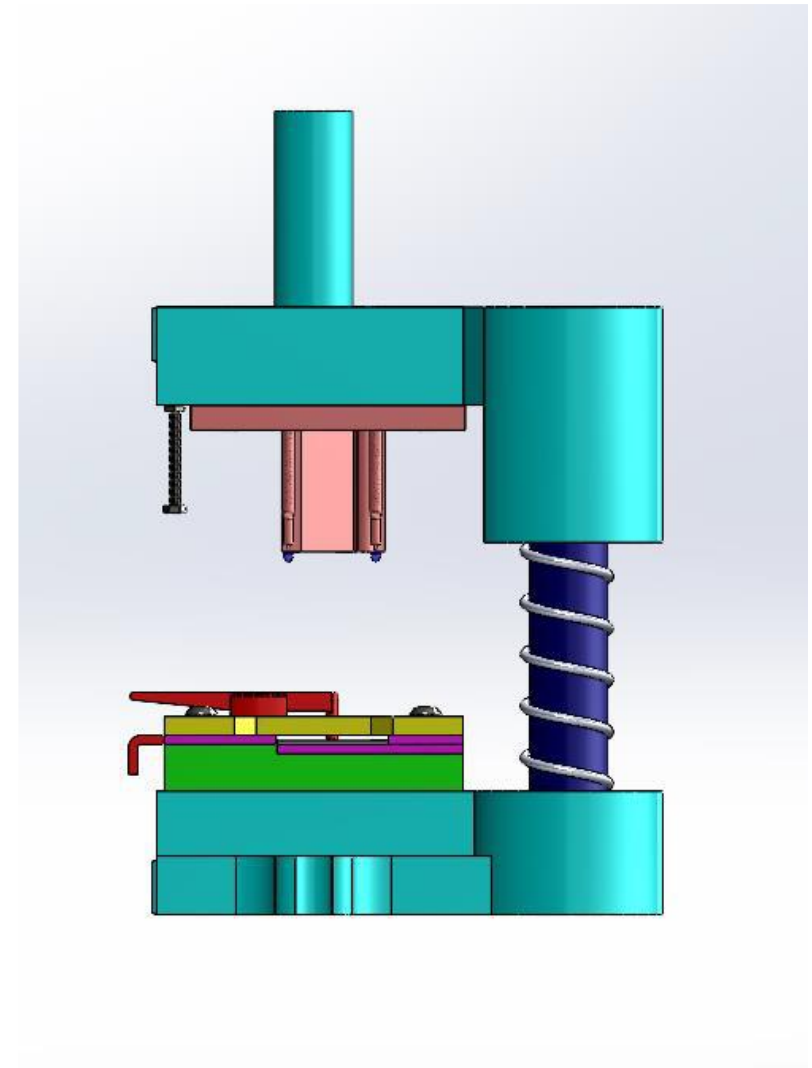
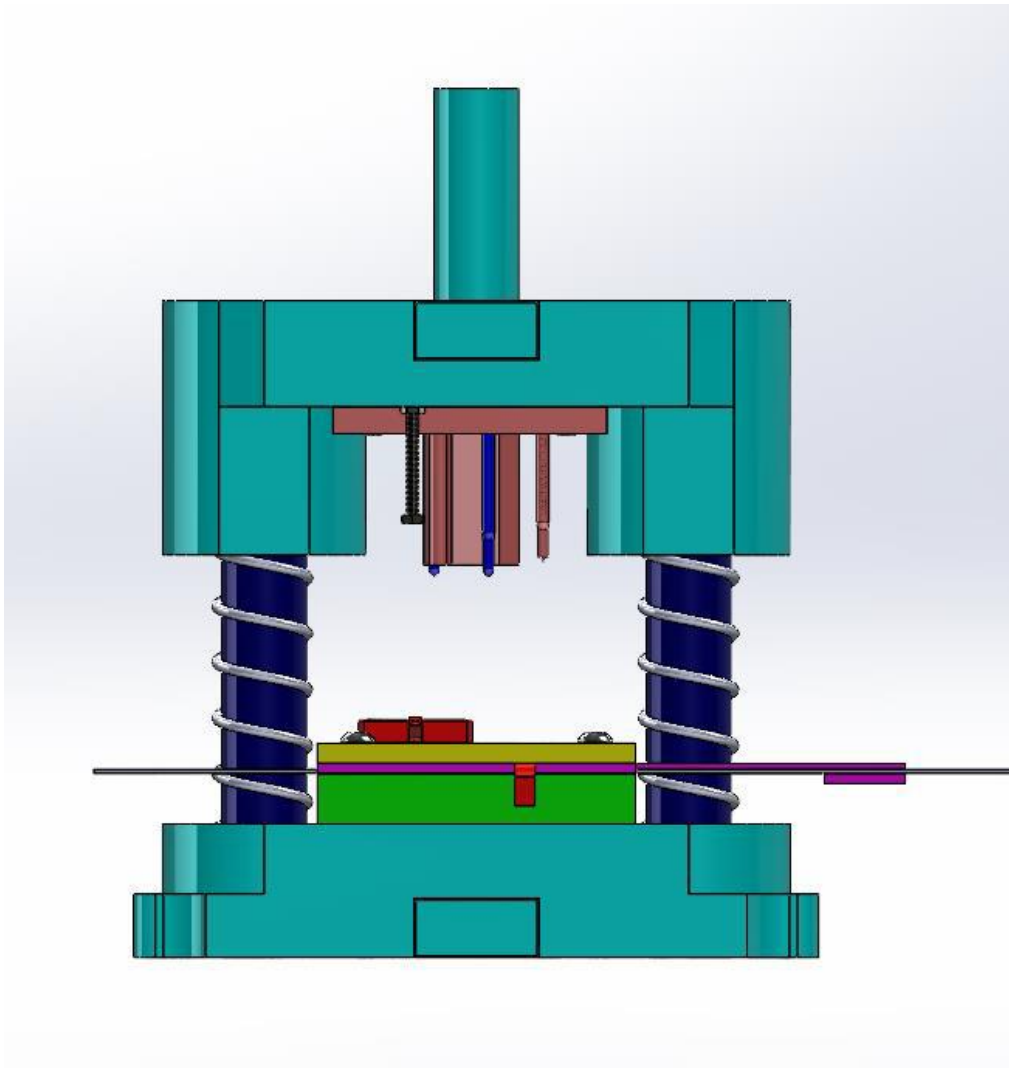
Parts/Strip: **38**
Total/Sheet: **798 pieces**

Die Assembly Overview

3D CAD REPRESENTATION

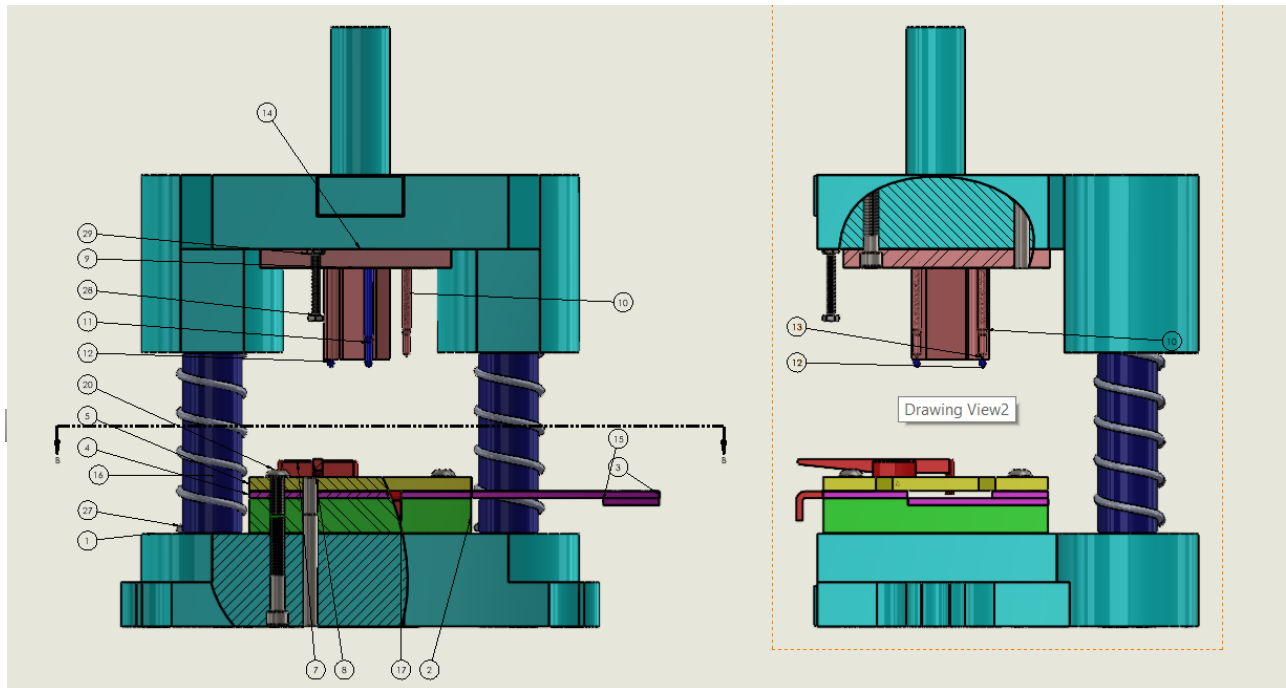
- **Die Shoes (Cyan):** Rear-pillar guide configuration.
- **Stripper Plate (Yellow):** Spring-actuated positive stripping.
- **Gages (Magenta):** Front and back strip pass line guides.
- **Stops (Red):** Finger stop & automatic trigger stop.
- **Die Block (Green):** Hardened matrix base.





Bill of Materials (BOM) & Section View

INVENTORY

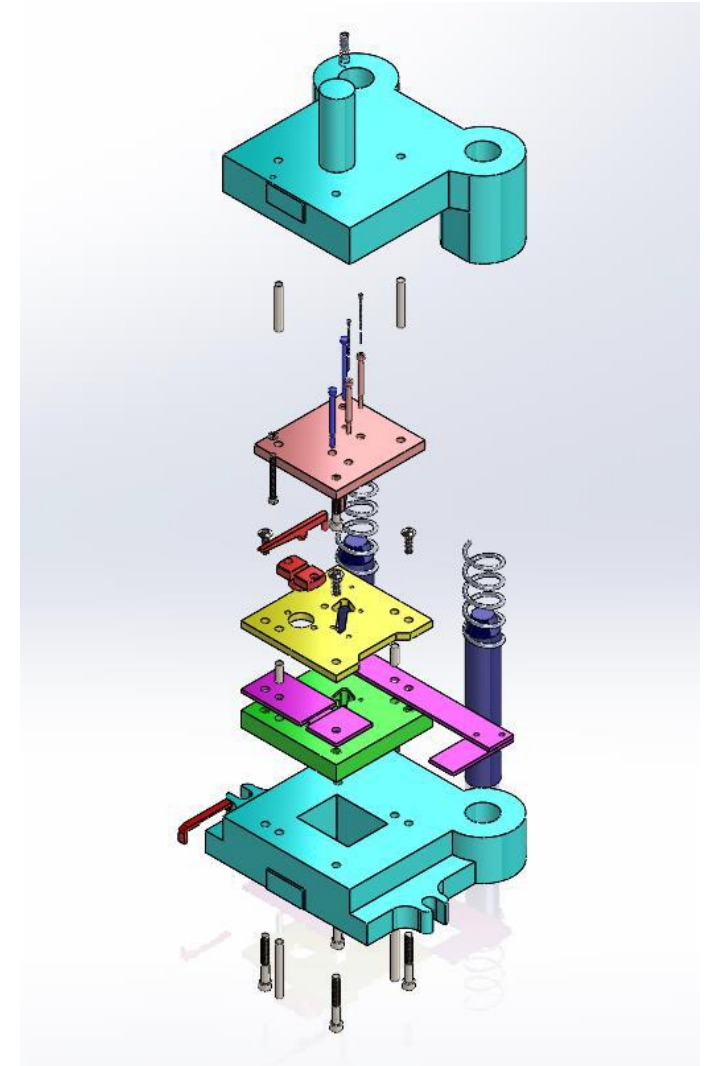


#	Part Name	Qty
1	Die Holder (Lower Shoe)	1
2	Die Block (Matrix)	1
3, 4	Gages (Front & Back)	2
5	Stripper Plate	1
7, 8	Automatic Stop System	2
9	Blanking Punch	1
10	Piercing Punch	2
11, 12	Pilots (Direct & Indirect)	2
13	Shedder Pin	2
14	Punch Holder (Upper Shoe)	1
16	Guide Post	2
17	Finger Stop	1
18-20	Socket Head Screws	10
21-25	Dowel Pins (ASME B18.8.2)	8
26-29	Springs & Trip Hardware	4

Exploded Assembly View

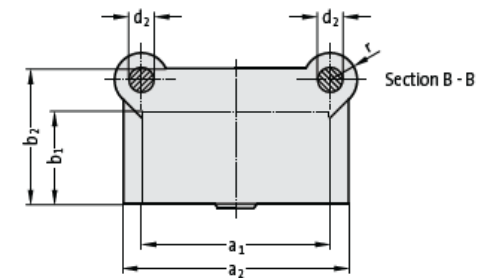
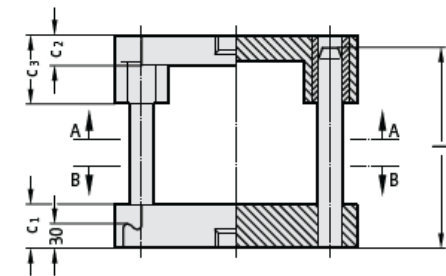
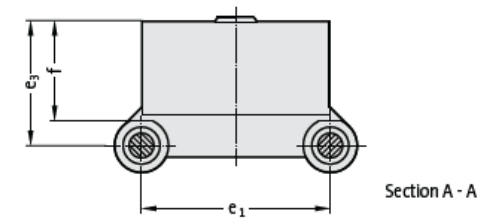
COMPONENT STACKING

- **Upper Subassembly:** Punch holder, blanking & piercing punches, pilots.
- **Intermediate Subassembly:** Stripper plate, stops, and stock gages.
- **Lower Subassembly:** Die block, strip rest, lower shoe, and guide posts.

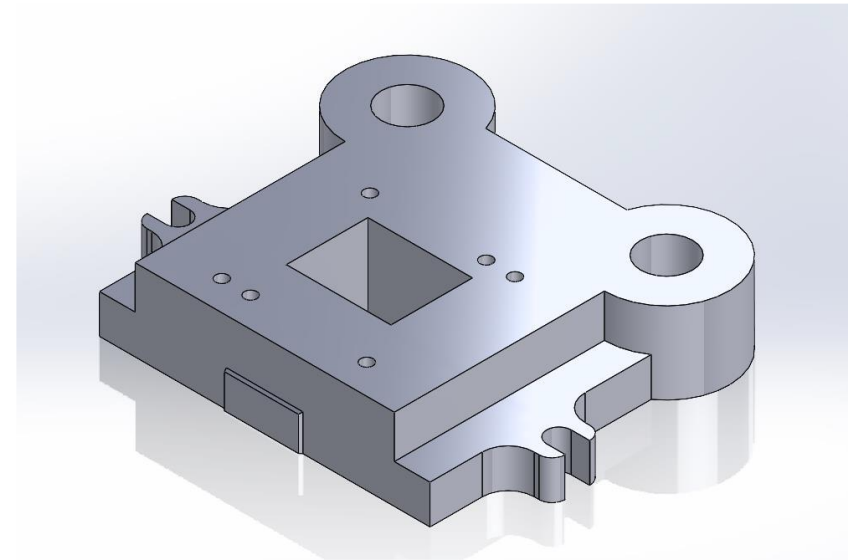
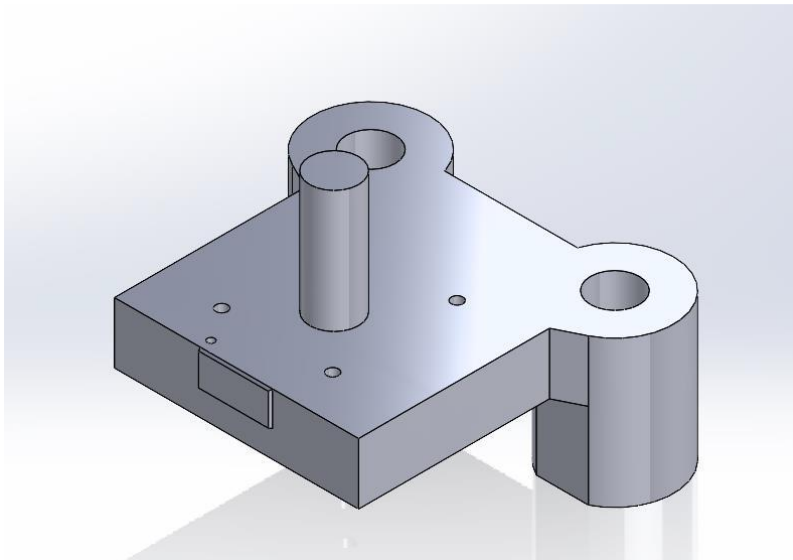


DIE SET DIN 9822 SHAPE C

201.31.



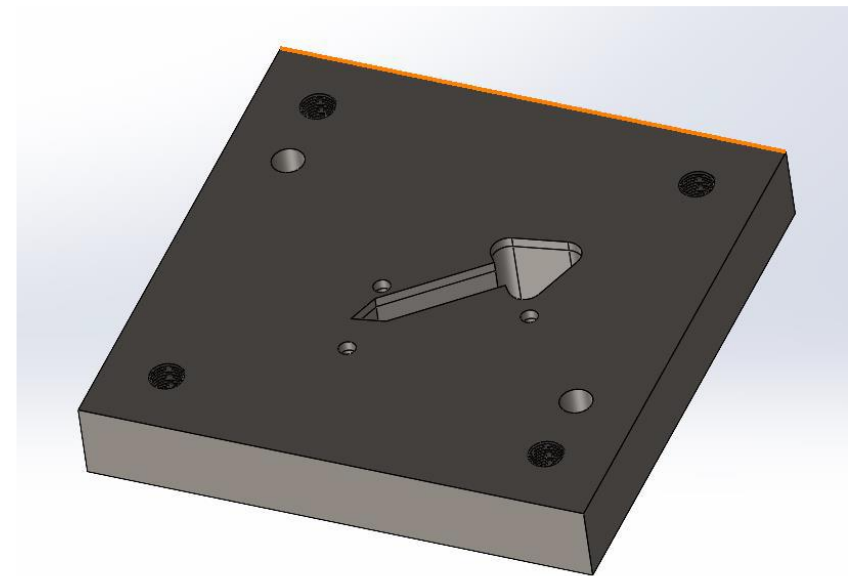
Selected from **FIBRO catalog (Series 201.31)** tool steel with rear guide pillars for unrestricted front feeding.



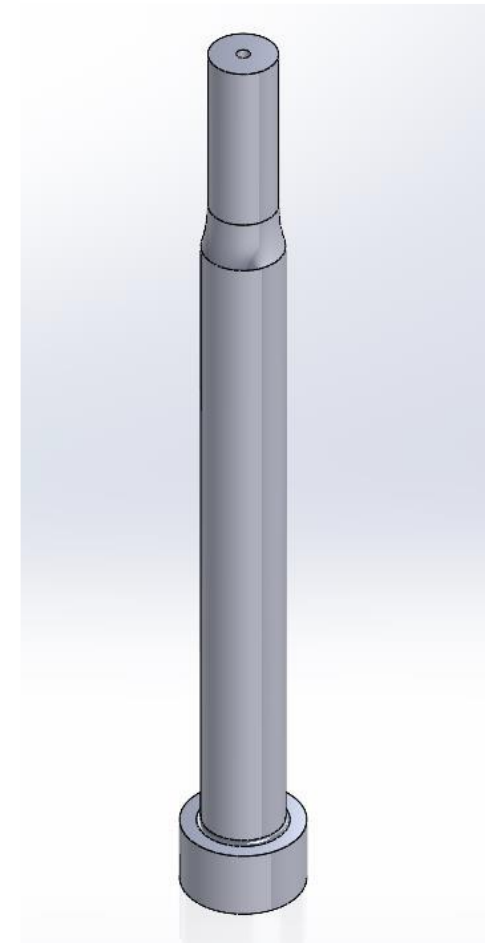
Die Block (Matrix) Design

CLEARANCES

Sized based on sheet thickness range and minimum edge margin C, featuring relief tapers for smooth slug ejection.



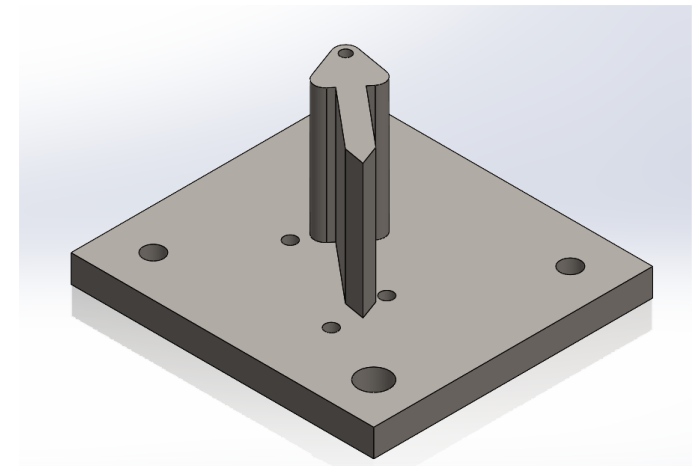
Stepped round punch with an integrated spring-loaded ejector (shedder) pin to prevent slug pulling during high-speed piercing.



Blanking Punch & Integrated Holder

RIGIDITY

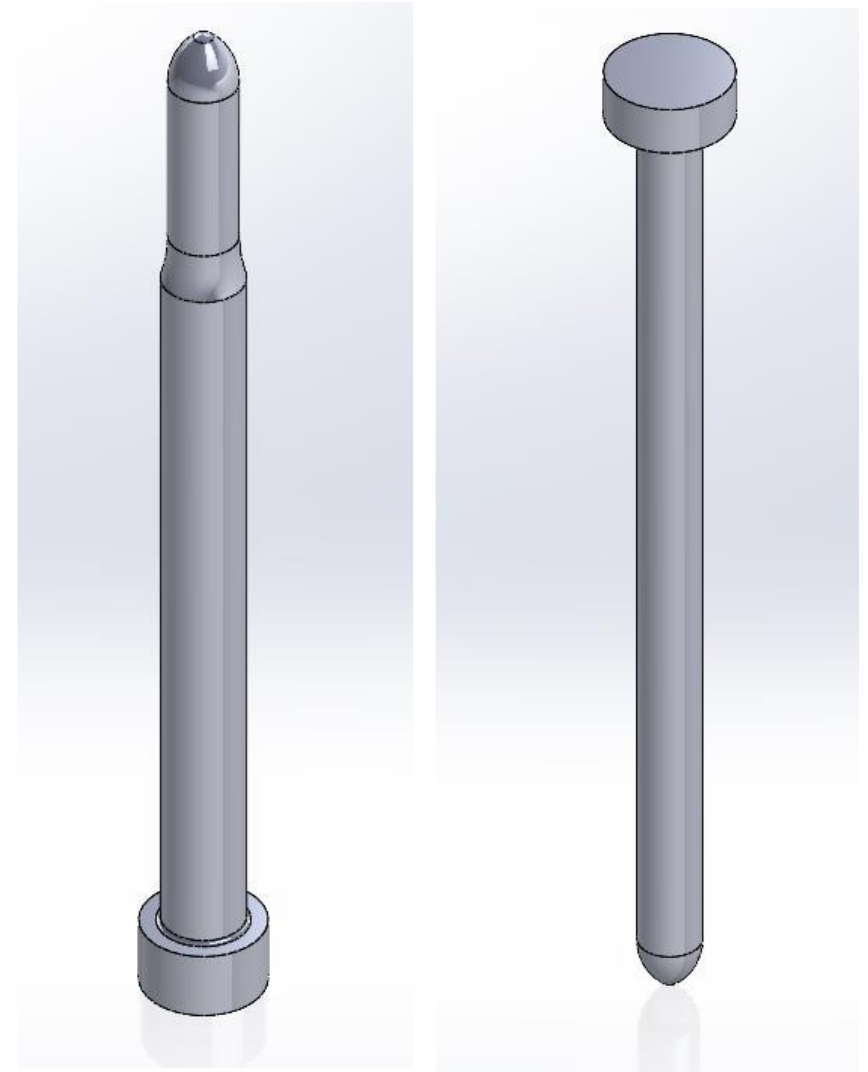
Machined as a single solid piece with its mounting flange to maximize rigidity and incorporate staged cutting action.



Progressive Pilot Pins

ALIGNMENT

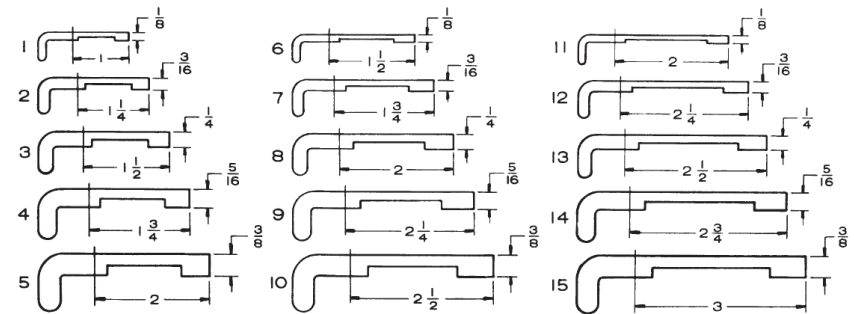
Direct pilot engages the Ø5.00 mm hole; indirect pilots correct strip feeding misalignments using a parabolic bullet-nose profile.



Manual Finger Stop (Initial Feed)

SIZE NO. 12

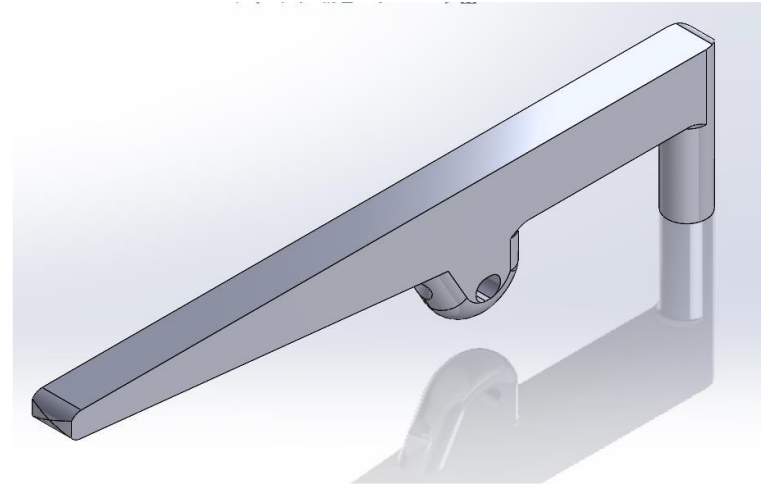
Includes a compression spring to serve simultaneously as an active stock pusher against the rear gage.



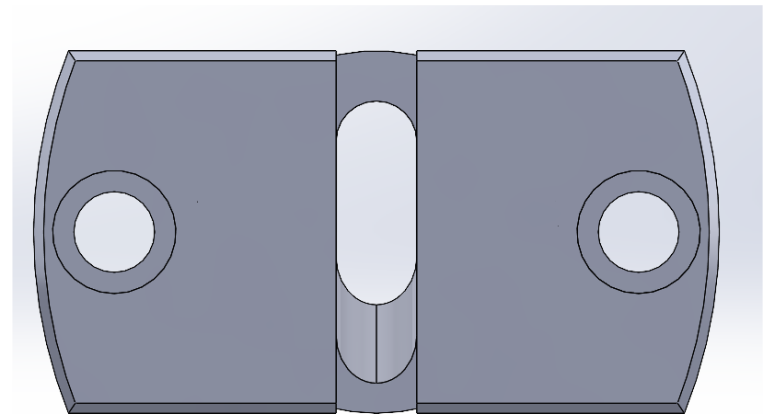
A	1 SMOOTH DIE HOLE CONTOUR		2 INSIDE CORNERS		3 SHARP INSIDE CORNERS	
	Front Spacer Width	Finger Stop No.	Front Spacer Width	Finger Stop No.	Front Spacer Width	Finger Stop No.
0 to 1/16	1	1	1 1/2	6	2	11
1/16 to 1/8	1 1/4	2	1 3/4	7	2 1/4	12
1/8 to 3/16	1 1/2	3	2	8	2 1/2	13
3/16 to 1/4	1 3/4	4	2 1/4	9	2 3/4	14
over 1/4	2	5	2 1/2	10	3	15

Automatic Trigger Stop Mechanism

MODEL NO. 2

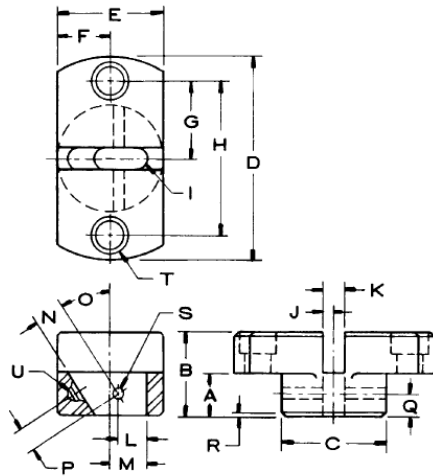


Actuated by an adjustable trip screw mounted on the upper die shoe for chatter-free progressive feeding control.



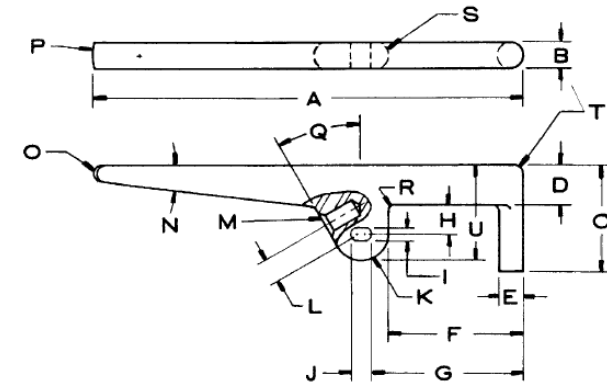
Automatic Stop Details

STANDARDS



NO.	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	1/4	21/32	1	2	1	1/2	3/4	1 1/2	1/8	1/8	1/4	5/16	23/64	5/16
2	3/8	13/16	1 1/8	2 1/8	1 1/8	9/16	13/16	1 5/8	1/8	1/8	1/4	11/32	13/32	11/32
3	1/2	31/32	1 1/4	2 3/8	1 1/4	5/8	29/32	1 13/16	1/8	1/8	1/4	11/32	7/16	3/8
4	5/8	1 1/8	1 3/8	2 1/2	1 3/8	11/16	31/32	1 15/16	5/32	5/32	5/16	13/32	1/2	7/16
5	3/4	1 9/32	1 1/2	2 7/8	1 1/2	3/4	1 3/32	2 3/16	5/32	5/32	5/16	15/32	9/16	15/32
6	7/8	1 7/16	1 5/8	3	1 5/8	13/16	1 5/32	2 5/16	5/32	5/32	5/16	1/2	5/8	17/32

NO.	O	P	Q	R	S	T	U
1	33°	7/32	1/8	1/32 x 45°	3/32	17/64 Dr.-13/32 C'bore 1/4 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.
2	33°	17/64	3/16	1/32 x 45°	1/8	17/64 Dr.-13/32 C'bore 1/4 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.
3	33°	17/64	1/4	1/32 x 45°	1/8	21/64 Dr.-15/32 C'bore 5/16 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.
4	33°	5/16	5/16	1/32 x 45°	3/16	21/64 Dr.-15/32 C'bore 5/16 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.
5	33°	11/32	3/8	1/32 x 45°	3/16	25/64 Dr.-19/32 C'bore 9/16 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.
6	33°	3/8	7/16	1/32 x 45°	3/16	25/64 Dr.-19/32 C'bore 9/16 Dp.	3/16 Dr. 3/32 Dp. 45° C'sink 3/64 Dp.

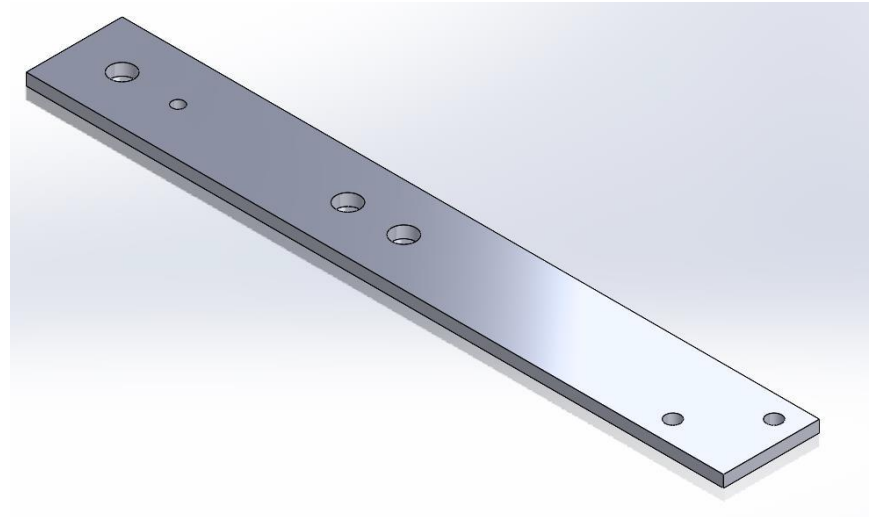


NO.	A	B	C	D	E	F	G	H	I	J	K	L
1	4	1/4	25/32	11/32	1/4	1 3/16	1 3/8	3/16	3/32	1/8	9/32 R.	3/16
2	4 3/16	1/4	15/16	3/8	1/4	1 1/4	1 7/16	1/4	1/8	5/32	9/32 R.	1/4
3	4 3/8	1/4	1 3/32	13/32	1/4	1 3/8	1 9/16	5/16	1/8	5/32	9/32 R.	1/4
4	4 9/16	5/16	1 1/4	7/16	5/16	1 15/32	1 11/16	3/8	3/16	7/32	11/32 R.	9/32
5	4 13/16	5/16	1 15/32	15/32	5/16	1 17/32	1 13/16	7/16	3/16	7/32	13/32 R.	5/16
6	5	5/16	1 11/16	1/2	5/16	1 5/8	1 15/16	1/2	3/16	7/32	7/16 R.	3/8

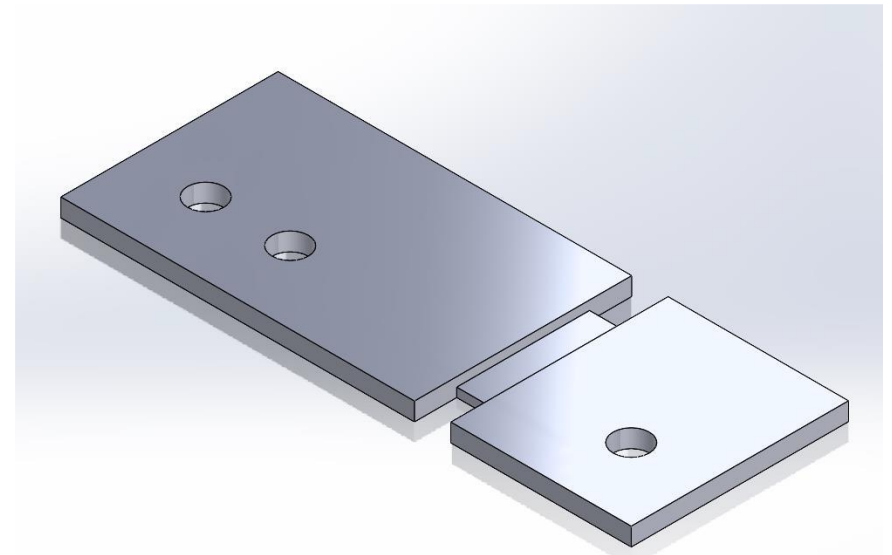
NO.	M	N	O	P	Q	R	S	T	U
1	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	6°	1/16 R.	1/2 R.	30°	3/64 R.	1/8 R.	1/16 R.	21/32
2	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	6°	5/64 R.	1/2 R.	30°	3/64 R.	1/8 R.	5/64 R.	13/16
3	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	6 1/2°	5/64 R.	1/2 R.	30°	3/64 R.	1/8 R.	3/32 R.	31/32
4	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	6 1/2°	3/32 R.	1/2 R.	30°	3/64 R.	5/32 R.	7/64 R.	1 1/8
5	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	7°	3/32 R.	1/2 R.	30°	3/64 R.	5/32 R.	1/8 R.	1 9/32
6	3/16 Dr. 3/8 Dp. 45° C/sink 1/16 Dp.	7 1/2°	3/32 R.	1/2 R.	30°	3/64 R.	5/32 R.	5/32 R.	1 7/16

Strip Guidance Gages

PASSLINE



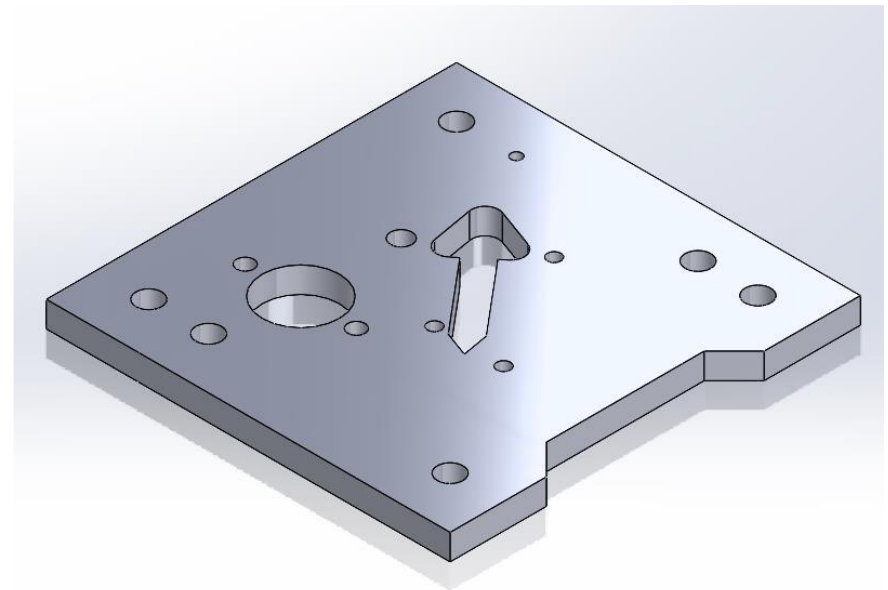
Front and back gages guide the strip along its feeding axis with proper clearance preventing jamming.



Fixed Stripper Plate

STRIPPING

Prevents sheet metal from adhering to punches on return stroke with precision clearance cutouts.



- **Socket Head Cap Screws:** 3/8 inch with 24 UNF unified fine thread.
- **Dowel Pins:** Hardened ASME B18.8.2 pins (0.3752" diameter) for concentricity.
- **Springs & Trip Screws:** Die post springs and M6 trip screw.

Single Part Area: **803.06 mm²** | Sheet Area: **1,200,000 mm²**

Parts per Sheet: **798 pieces**

Material Utilization Ratio: **53.4%**

Thank You!

Questions & Discussion

Designer: **Arian Mahjoubi**

Student ID: **401107443**